

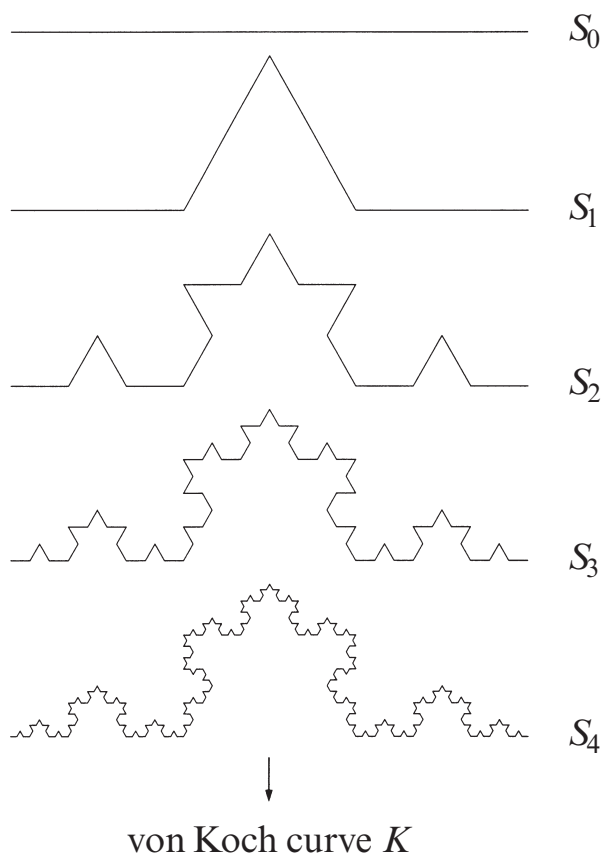


Figure 11.3.1

A Paradox

What is the dimension of the von Koch curve? Since it's a curve, you might be tempted to say it's one-dimensional. But the trouble is that K has *infinite arc length*! To see this, observe that if the length of S_0 is L_0 , then the length of S_1 is $L_1 = \frac{4}{3}L_0$, because S_1 contains four segments, each of length $\frac{1}{3}L_0$. The length increases by a factor of $\frac{4}{3}$ at each stage of the construction, so $L_n = (\frac{4}{3})^n L_0 \rightarrow \infty$ as $n \rightarrow \infty$.

Moreover, the arc length between *any* two points on K is infinite, by similar reasoning. Hence points on K aren't determined by their arc length from a particular point, because every point is infinitely far from every other!

This suggests that K is more than one-dimensional. But would we really want to say that K is two-dimensional? It certainly doesn't seem to have any "area." So the dimension should be *between* 1 and 2, whatever that means.

With this paradox as motivation, we now consider some improved notions of dimension that can cope with fractals.

Similarity Dimension

The simplest fractals are self-similar, i.e., they are made of scaled-down copies of themselves, all the way down to arbitrarily small scales. The dimension of such fractals can be defined by extending an elementary observation about *classical* self-similar sets like line segments, squares, or cubes. For instance, consider the square region shown in Figure 11.3.2.

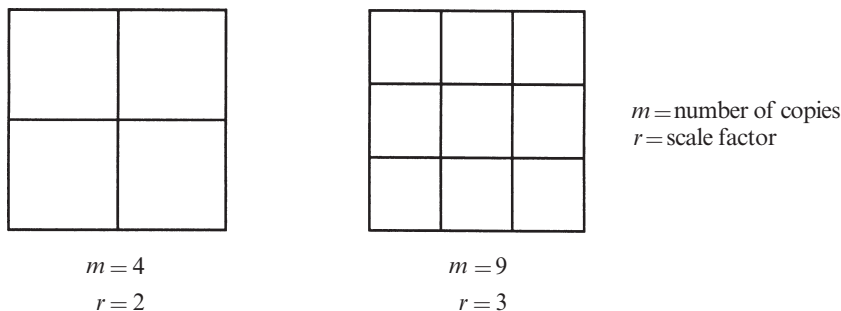


Figure 11.3.2

If we shrink the square by a factor of 2 in each direction, it takes four of the small squares to equal the whole. Or if we scale the original square down by a factor of 3, then nine small squares are required. In general, if we reduce the linear dimensions of the square region by a factor of r , it takes r^2 of the smaller squares to equal the original.

Now suppose we play the same game with a solid cube. The results are different: if we scale the cube down by a factor of 2, it takes eight of the smaller cubes to make up the original. In general, if the cube is scaled down by r , we need r^3 of the smaller cubes to make up the larger one.

The exponents 2 and 3 are no accident; they reflect the two-dimensionality of the square and the three-dimensionality of the cube. This connection between dimensions and exponents suggests the following definition. Suppose that a self-similar set is composed of m copies of itself scaled down by a factor of r . Then the **similarity dimension** d is the exponent defined by $m = r^d$, or equivalently,

$$d = \frac{\ln m}{\ln r}.$$

This formula is easy to use, since m and r are usually clear from inspection.

EXAMPLE 11.3.1:

Find the similarity dimension of the Cantor set C .

Solution: As shown in Figure 11.3.3, C is composed of two copies of itself, each scaled down by a factor of 3.

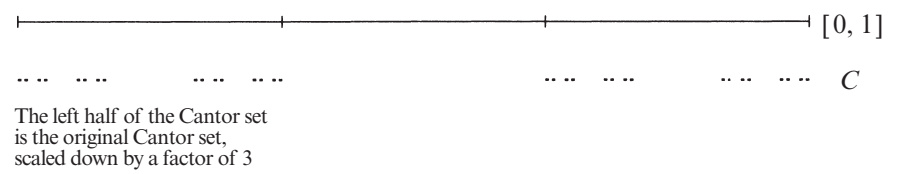


Figure 11.3.3

So $m = 2$ when $r = 3$. Therefore $d = \ln 2 / \ln 3 \approx 0.63$. ■

In the next example we confirm our earlier intuition that the von Koch curve should have a dimension between 1 and 2.

EXAMPLE 11.3.2:

Show that the von Koch curve has a similarity dimension of $\ln 4 / \ln 3 \approx 1.26$.

Solution: The curve is made up of four equal pieces, each of which is similar to the original curve but is scaled down by a factor of 3 in both directions. One of these pieces is indicated by the arrows in Figure 11.3.4.

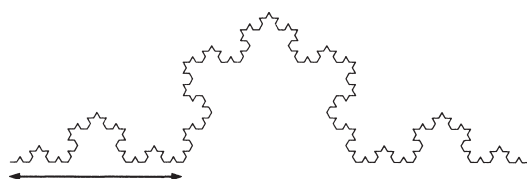


Figure 11.3.4

Hence $m = 4$ when $r = 3$, and therefore $d = \ln 4 / \ln 3$. ■

More General Cantor Sets

Other self-similar fractals can be generated by changing the recursive procedure. For instance, to obtain a new kind of Cantor set, divide an interval into five equal pieces, delete the second and fourth subintervals, and then repeat this process indefinitely (Figure 11.3.5).

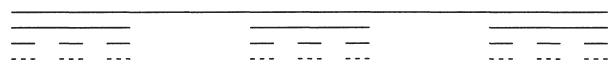


Figure 11.3.5

We call the limiting set the *even-fifths Cantor set*, since the even fifths are removed at each stage. (Similarly, the standard Cantor set of Section 11.2 is often called the *middle-thirds Cantor set*.)

EXAMPLE 11.3.3:

Find the similarity dimension of the even-fifths Cantor set.

Solution: Let the original interval be denoted S_0 , and let S_n denote the n th stage of the construction. If we scale S_n down by a factor of five, we get one third of the set S_{n+1} . Now setting $n = \infty$, we see that the even-fifths Cantor set S_∞ is made of three copies of itself, shrunk by a factor of 5. Hence $m = 3$ when $r = 5$, and so $d = \ln 3 / \ln 5$. ■

There are so many different Cantor-like sets that mathematicians have abstracted their essence in the following definition. A closed set S is called a **topological Cantor set** if it satisfies the following properties:

1. S is “totally disconnected.” This means that S contains no connected subsets (other than single points). In this sense, all points in S are separated from each other. For the middle-thirds Cantor set and other subsets of the real line, this condition simply says that S contains no intervals.
2. On the other hand, S contains no “isolated points.” This means that every point in S has a neighbor arbitrarily close by—given any point $p \in S$ and any small distance $\varepsilon > 0$, there is some other point $q \in S$ within a distance ε of p .

The paradoxical aspects of Cantor sets arise because the first property says that points in S are spread apart, whereas the second property says they’re packed together! In Exercise 11.3.6, you’re asked to check that the middle-thirds Cantor set satisfies both properties.

Notice that the definition says nothing about self-similarity or dimension. These notions are geometric rather than topological; they depend on concepts of distance, volume, and so on, which are too rigid for some purposes. Topological features are more robust than geometric ones. For instance, if we continuously deform a self-similar Cantor set, we can easily destroy its self-similarity but properties 1 and 2 will persist. When we study strange attractors in Chapter 12, we’ll see that the cross sections of strange attractors are often topological Cantor sets, although they are not necessarily self-similar.

11.4 Box Dimension

To deal with fractals that are not self-similar, we need to generalize our notion of dimension still further. Various definitions have been proposed; see Falconer (1990) for a lucid discussion. All the definitions share the idea of “measurement at a scale ε ”—roughly speaking, we measure the set in a way that ignores irregularities of size less than ε , and then study how the measurements vary as $\varepsilon \rightarrow 0$.

Definition of Box Dimension

One kind of measurement involves covering the set with boxes of size ε (Figure 11.4.1).

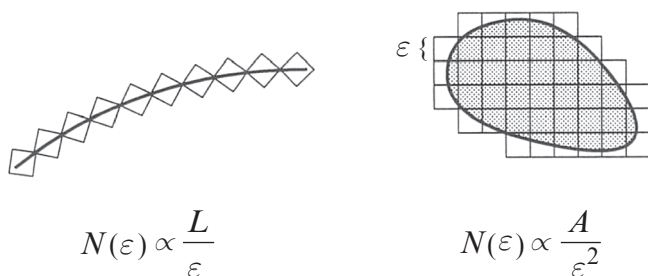


Figure 11.4.1

Let S be a subset of D -dimensional Euclidean space, and let $N(\varepsilon)$ be the minimum number of D -dimensional cubes of side ε needed to cover S . How does $N(\varepsilon)$ depend on ε ? To get some intuition, consider the classical sets shown in Figure 11.4.1. For a smooth curve of length L , $N(\varepsilon) \propto L/\varepsilon$; for a planar region of area A bounded by a smooth curve, $N(\varepsilon) \propto A/\varepsilon^2$. The key observation is that the dimension of the set equals the exponent d in the **power law** $N(\varepsilon) \propto 1/\varepsilon^d$.

This power law also holds for most fractal sets S , except that d is no longer an integer. By analogy with the classical case, we interpret d as a dimension, usually called the *capacity* or **box dimension** of S . An equivalent definition is

$$d = \lim_{\varepsilon \rightarrow 0} \frac{\ln N(\varepsilon)}{\ln(1/\varepsilon)}, \text{ if the limit exists.}$$

EXAMPLE 11.4.1:

Find the box dimension of the Cantor set.

Solution: Recall that the Cantor set is covered by each of the sets S_n used in its construction (Figure 11.2.1). Each S_n consists of 2^n intervals of length $(1/3)^n$, so if we pick $\varepsilon = (1/3)^n$, we need all 2^n of these intervals to cover the Cantor set. Hence $N = 2^n$ when $\varepsilon = (1/3)^n$. Since $\varepsilon \rightarrow 0$ as $n \rightarrow \infty$, we find

$$d = \lim_{\varepsilon \rightarrow 0} \frac{\ln N(\varepsilon)}{\ln(1/\varepsilon)} = \frac{\ln(2^n)}{\ln(3^n)} = \frac{n \ln 2}{n \ln 3} = \frac{\ln 2}{\ln 3}$$

in agreement with the similarity dimension found in Example 11.3.1. ■

This solution illustrates a helpful trick. We used a discrete sequence $\varepsilon = (1/3)^n$ that tends to zero as $n \rightarrow \infty$, even though the definition of box dimension says that we should let $\varepsilon \rightarrow 0$ continuously. If $\varepsilon \neq (1/3)^n$, the covering will be slightly wasteful—some boxes hang over the edge of the set—but the limiting value of d is the same.

EXAMPLE 11.4.2:

A fractal that is *not* self-similar is constructed as follows. A square region is divided into nine equal squares, and then one of the small squares is selected at random and discarded. Then the process is repeated on each of the eight remaining small squares, and so on. What is the box dimension of the limiting set?

Solution: Figure 11.4.2 shows the first two stages in a typical realization of this random construction.

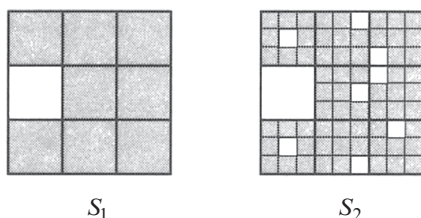


Figure 11.4.2

Pick the unit of length to equal the side of the original square. Then S_1 is covered (with no wastage) by $N = 8$ squares of side $\varepsilon = \frac{1}{3}$. Similarly, S_2 is covered by $N = 8^2$ squares of side $\varepsilon = (\frac{1}{3})^2$. In general, $N = 8^n$ when $\varepsilon = (\frac{1}{3})^n$. Hence

$$d = \lim_{\varepsilon \rightarrow 0} \frac{\ln N(\varepsilon)}{\ln(1/\varepsilon)} = \frac{\ln(8^n)}{\ln(3^n)} = \frac{n \ln 8}{n \ln 3} = \frac{\ln 8}{\ln 3}. \blacksquare$$

Critique of Box Dimension

When computing the box dimension, it is not always easy to find a minimal cover. There's an equivalent way to compute the box dimension that avoids this problem. We cover the set with a square mesh of boxes of side ε , count the number of occupied boxes $N(\varepsilon)$, and then compute d as before.

Even with this improvement, the box dimension is rarely used in practice. Its computation requires too much storage space and computer time, compared to other

types of fractal dimension (see below). The box dimension also suffers from some mathematical drawbacks. For example, its value is not always what it should be: the set of rational numbers between 0 and 1 can be proven to have a box dimension of 1 (Falconer 1990, p. 44), even though the set has only countably many points.

Falconer (1990) discusses other fractal dimensions, the most important of which is the *Hausdorff dimension*. It is more subtle than the box dimension. The main conceptual difference is that the Hausdorff dimension uses coverings by small sets of *varying* sizes, not just boxes of fixed size ε . It has nicer mathematical properties than the box dimension, but unfortunately it is even harder to compute numerically.

11.5 Pointwise and Correlation Dimensions

Now it's time to return to dynamics. Suppose that we're studying a chaotic system that settles down to a strange attractor in phase space. Given that strange attractors typically have fractal microstructure (as we'll see in Chapter 12), how could we estimate the fractal dimension?

First we generate a set of very many points $\{\mathbf{x}_i, i = 1, \dots, n\}$ on the attractor by letting the system evolve for a long time (after taking care to discard the initial transient, as usual). To get better statistics, we could repeat this procedure for several different trajectories. In practice, however, almost all trajectories on a strange attractor have the same long-term statistics so it's sufficient to run one trajectory for an extremely long time. Now that we have many points on the attractor, we could try computing the box dimension, but that approach is impractical, as mentioned earlier.

Grassberger and Procaccia (1983) proposed a more efficient approach that has become standard. Fix a point $\mathbf{x}$ on the attractor A . Let $N_\varepsilon(\varepsilon)$ denote the number of points on A inside a ball of radius ε about $\mathbf{x}$ (Figure 11.5.1).

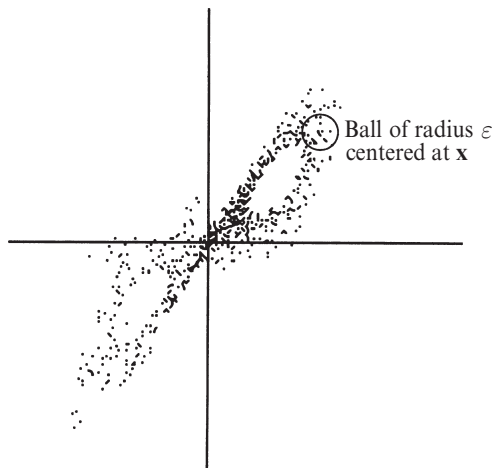


Figure 11.5.1

Most of the points in the ball are unrelated to the immediate portion of the trajectory through $\mathbf{x}$; instead they come from later parts that just happen to pass close to $\mathbf{x}$. Thus $N_{\mathbf{x}}(\varepsilon)$ measures how frequently a typical trajectory visits an ε neighborhood of $\mathbf{x}$.

Now vary ε . As ε increases, the number of points in the ball typically grows as a power law:

$$N_{\mathbf{x}}(\varepsilon) \propto \varepsilon^d,$$

where d is called the **pointwise dimension** at $\mathbf{x}$. The pointwise dimension can depend significantly on $\mathbf{x}$; it will be smaller in rarefied regions of the attractor. To get an overall dimension of A , one averages $N_{\mathbf{x}}(\varepsilon)$ over many $\mathbf{x}$. The resulting quantity $C(\varepsilon)$ is found empirically to scale as

$$C(\varepsilon) \propto \varepsilon^d,$$

where d is called the **correlation dimension**.

The correlation dimension takes account of the density of points on the attractor, and thus differs from the box dimension, which weights all occupied boxes equally, no matter how many points they contain. (Mathematically speaking, the correlation dimension involves an invariant measure supported on a fractal, not just the fractal itself.) In general, $d_{\text{correlation}} \leq d_{\text{box}}$, although they are usually very close (Grassberger and Procaccia 1983).

To estimate d , one plots $\log C(\varepsilon)$ vs. $\log \varepsilon$. If the relation $C(\varepsilon) \propto \varepsilon^d$ were valid for all ε , we'd find a straight line of slope d . In practice, the power law holds only over an intermediate range of ε (Figure 11.5.2).

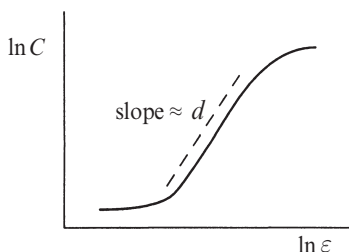


Figure 11.5.2

The curve saturates at large ε because the ε -balls engulf the whole attractor and so $N_{\mathbf{x}}(\varepsilon)$ can grow no further. On the other hand, at extremely small ε , the only point in each ε -ball is $\mathbf{x}$ itself. So the power law is expected to hold only in the *scaling region* where

(minimum separation of points on A) $\ll \varepsilon \ll$ (diameter of A).